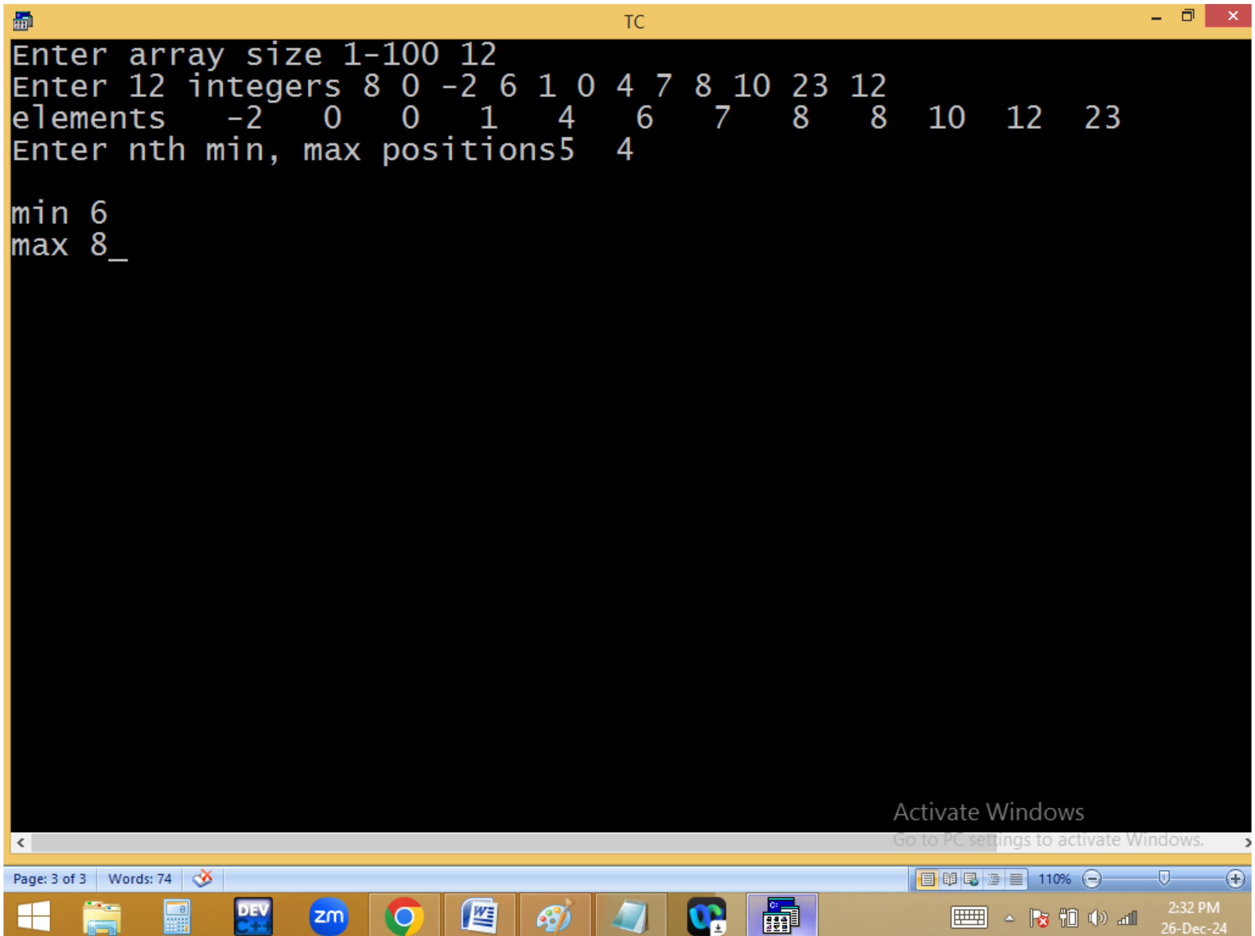
Finding nth max, nth min array elements:

```
#include<stdio.h>

#include<conio.h>

void main()

{

int a[100],n, t,i,j,max, min;clrscr();

printf("Enter array size 1-100 "); scanf("%d",&n);

printf("Enter %d integers ", n);

for(i=0;i<n;i++)scanf("%d",&a[i]);

for( i=0;i<=n-2;i++)

{

for( j=0;j<=n-i-2;j++)

{if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}}

}

printf("elements ");

for(i=0;i<n;i++)printf("%4d",a[i]);

printf("\nEnter nth min, max positions");

scanf("%d%d",&min,&max);
```

```
if(min==1)printf("1st min=%d\n",a[0]);

else

{

for(i=1;i<n;i++)

{

if(a[i]>a[i-1])

{min--;if(min==1){printf("\nmin %d\n",a[i]);break;}}

}

}

if(max==1)printf("1st max=%d",a[n-1]);

else

{

for(i=n-2;i>=0;i--)

{

if(a[i]<a[i+1])

{

max--;if(max==1){printf("max %d",a[i]);break;}}

}
```

```
}
getch();
}
```




```
if(min==1)p(a[0]); else

for( i=1 ; i<9;i++)
{
if(a[i]>a[i-1])min--;
if(min==1){p(a[i]);break;}
}

if(max==1)p(a[n-1]);
else
{
for(i=n-2;i>=0;i--)
{ if(a[i]<a[i-1])max--;
if(max==1)p(a[i]);break;
}
```

2	5	6	9	10	15	21	24	30
0	1	2	3	4	5	6	7	8

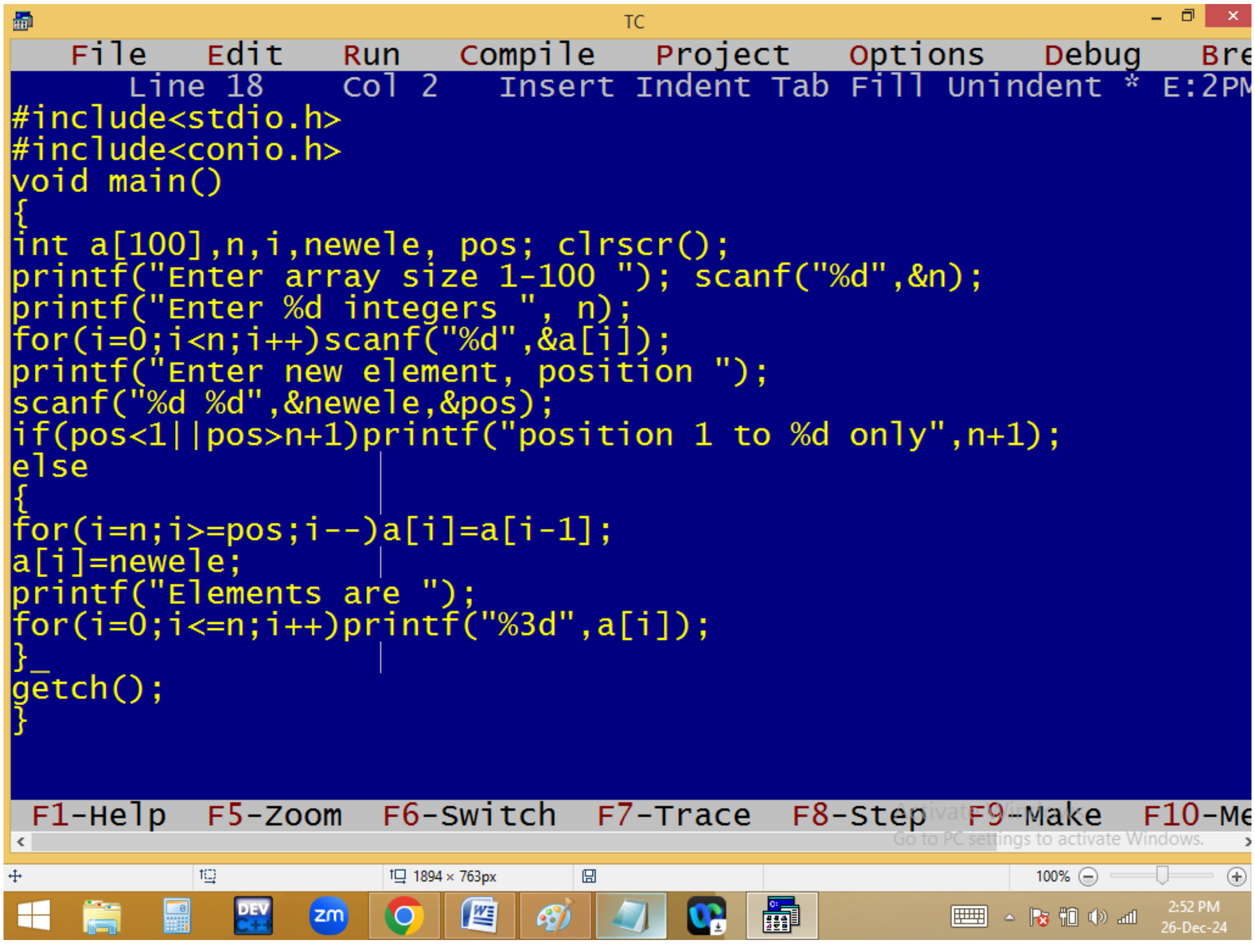
n	i
9	1
	2

~~3~~rd min
~~2~~ 6✓
1

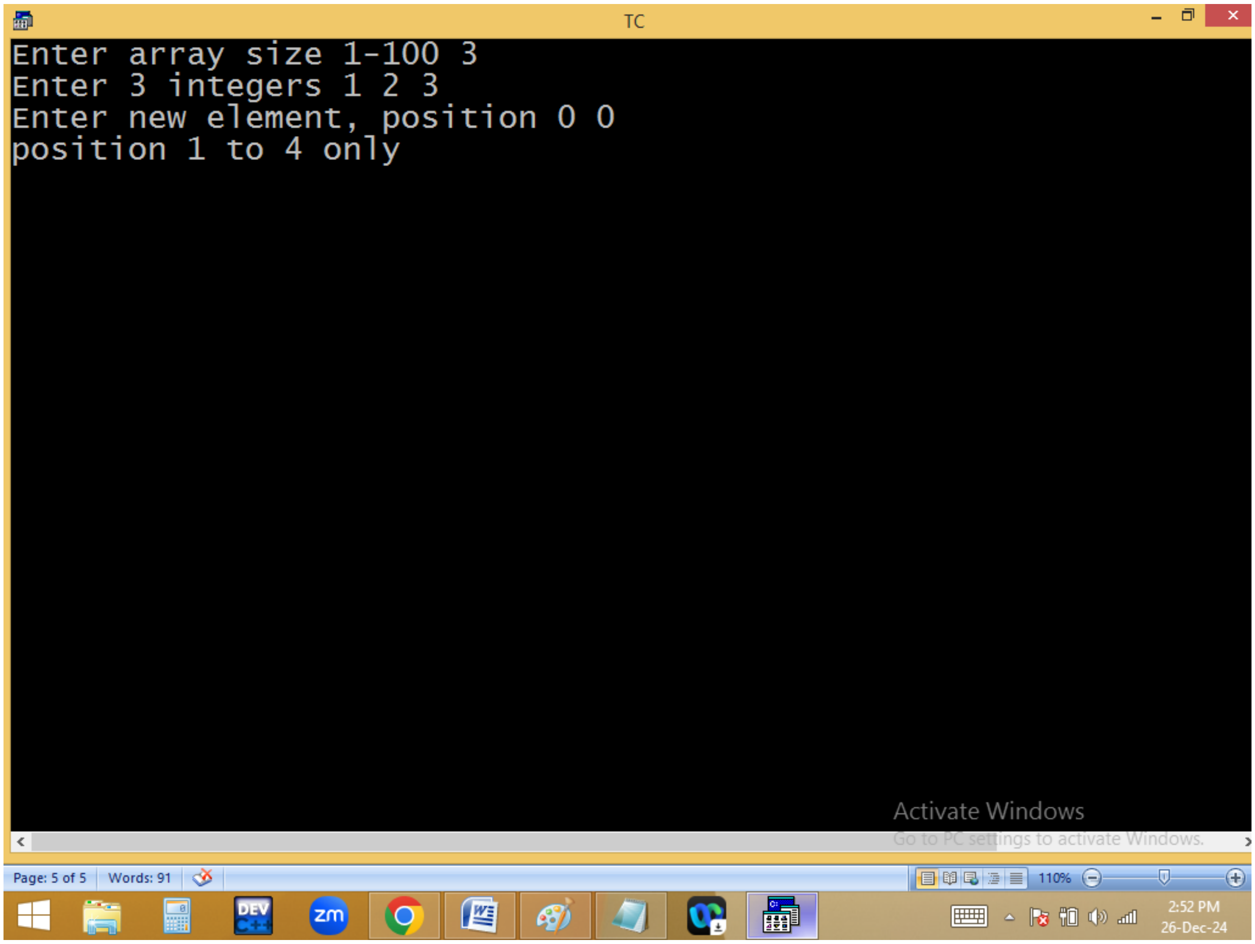
~~5~~th max
~~4~~ 10✓
~~3~~
~~2~~
1

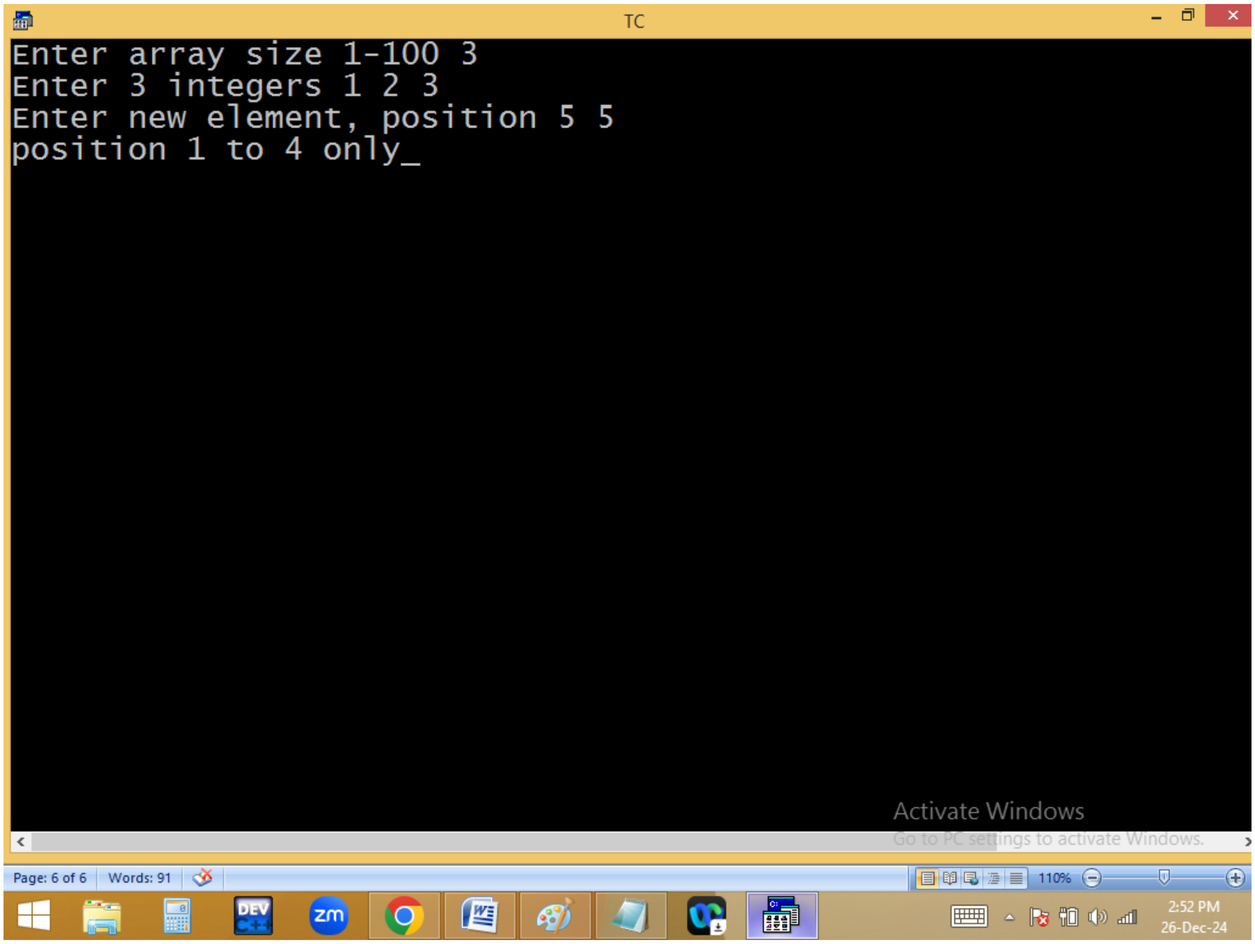
i
7
6
5
4✓

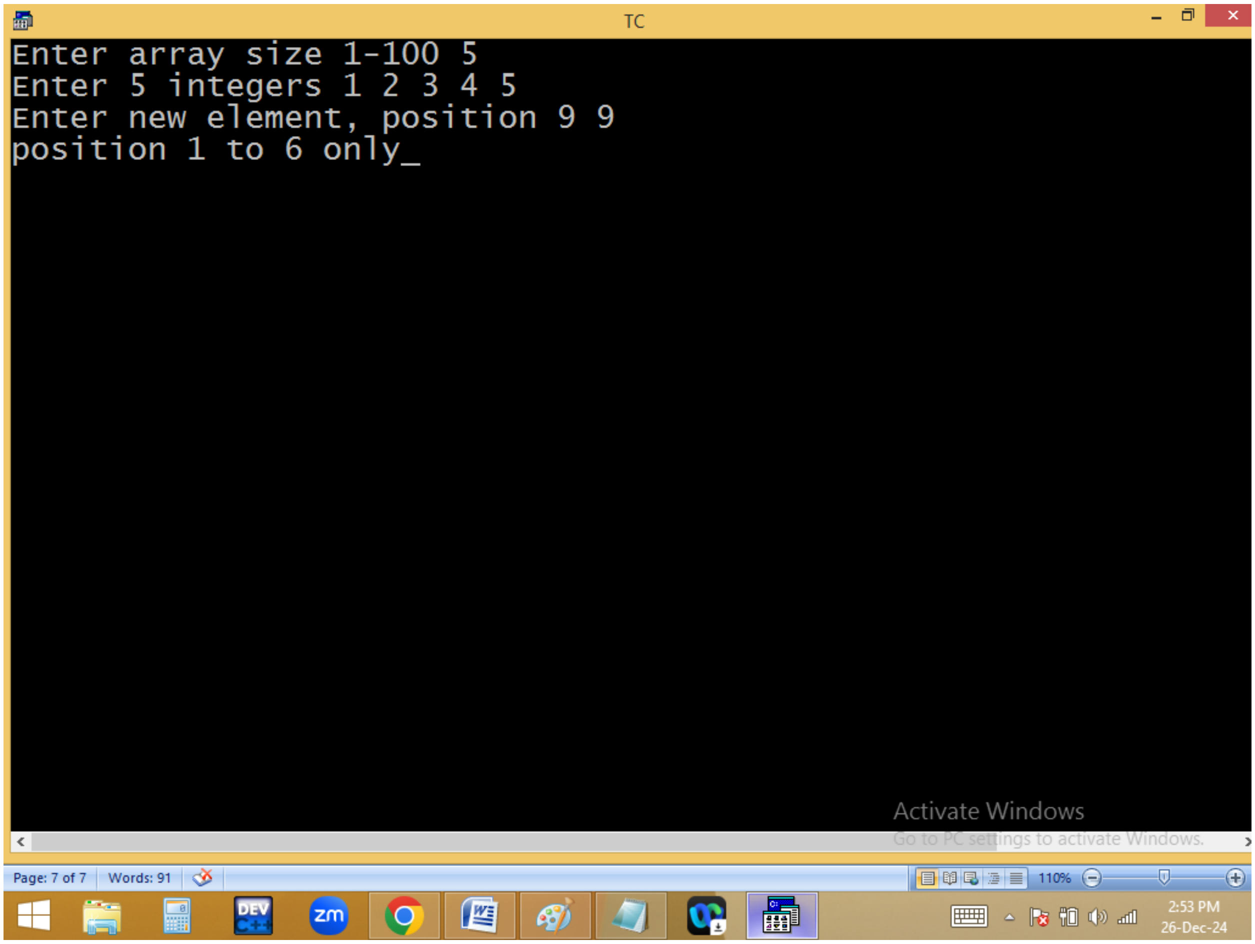
Inserting a new element in specified position of array [push – right shifting or array elements]:

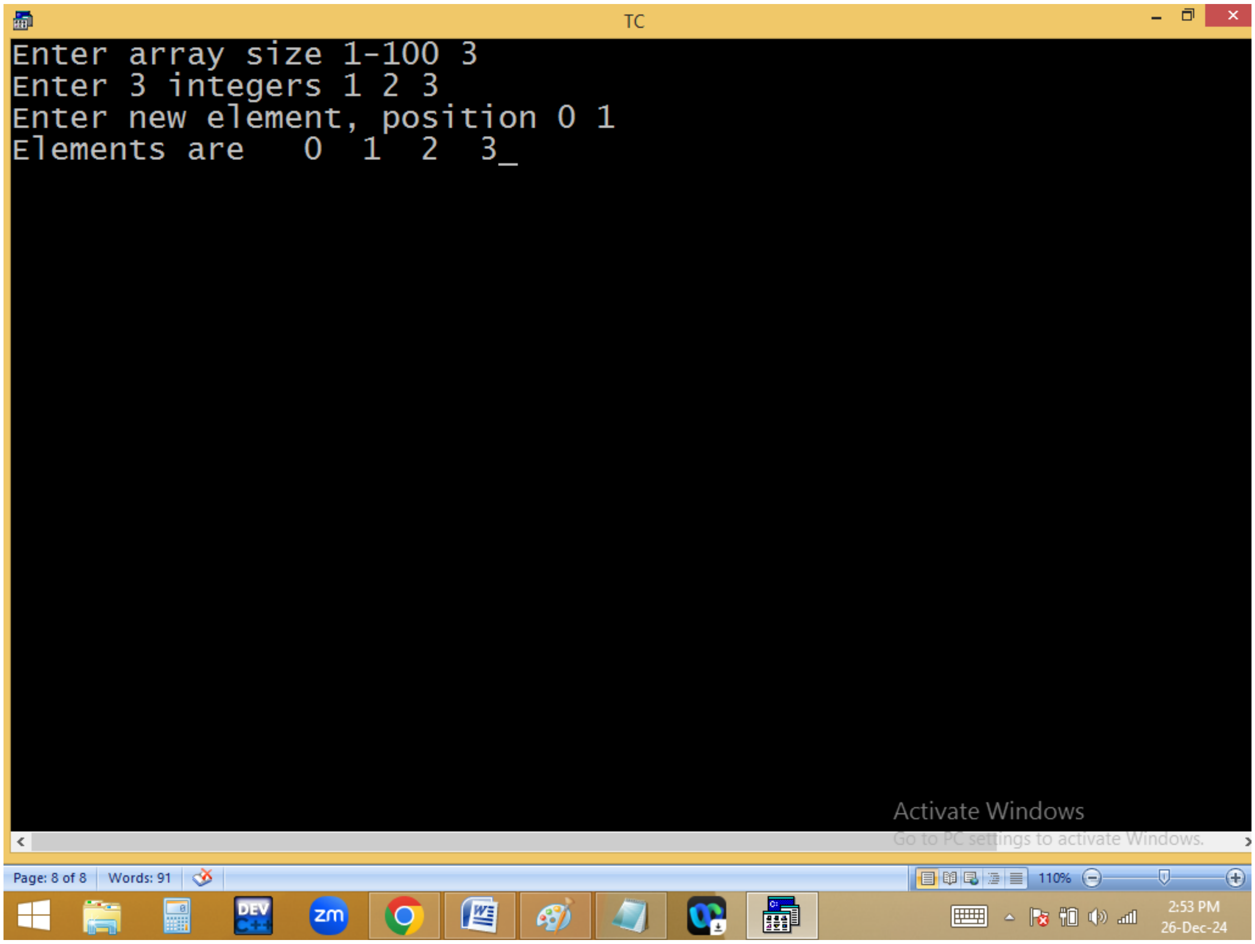


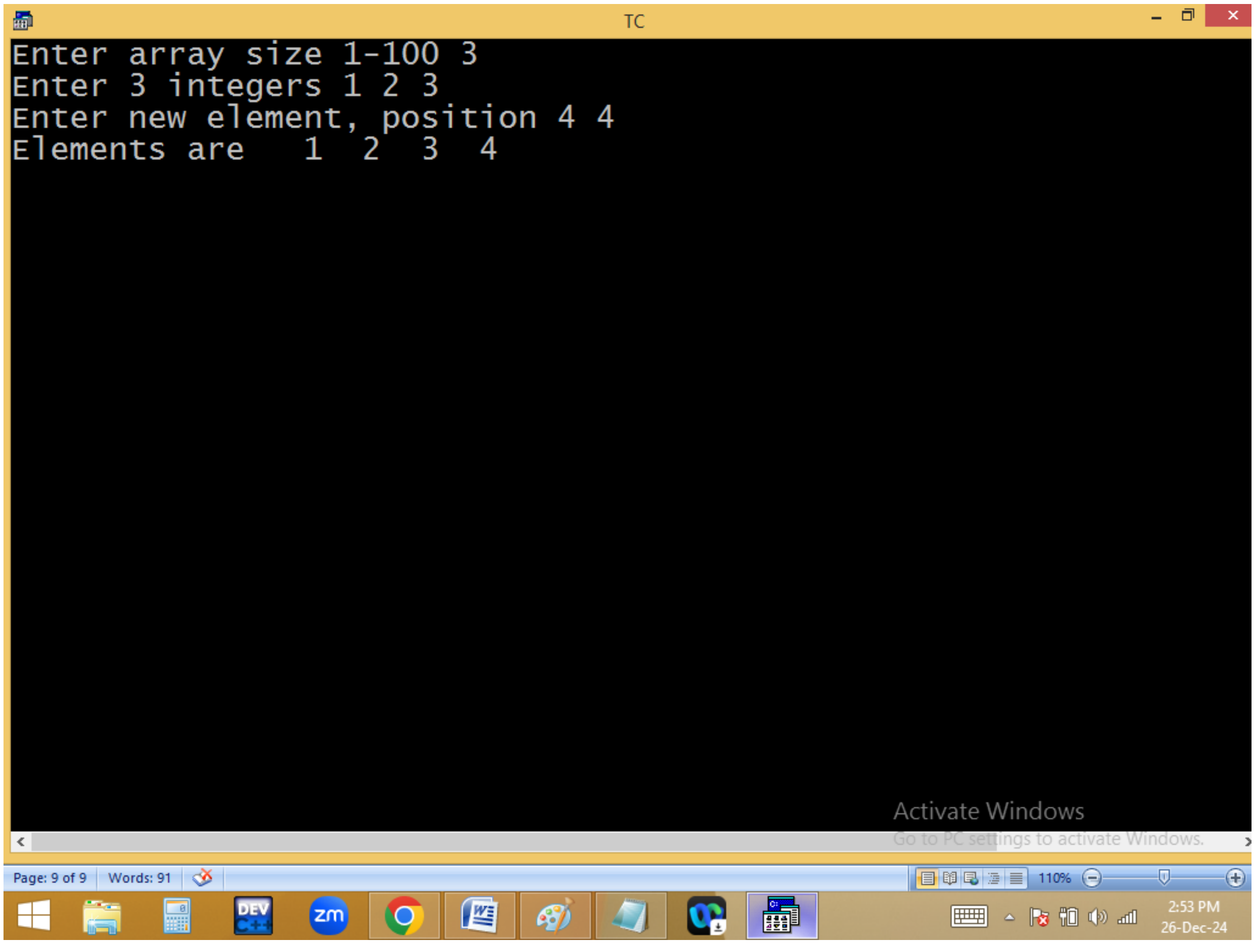

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,newele, pos; clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter new element, position ");
scanf("%d %d",&newele,&pos);
if(pos<1||pos>n+1)printf("position 1 to %d only",n+1);
else
{
for(i=n;i>=pos;i--)a[i]=a[i-1];
a[i]=newele;
printf("Elements are ");
for(i=0;i<=n;i++)printf("%3d",a[i]);
}
getch();
}
```

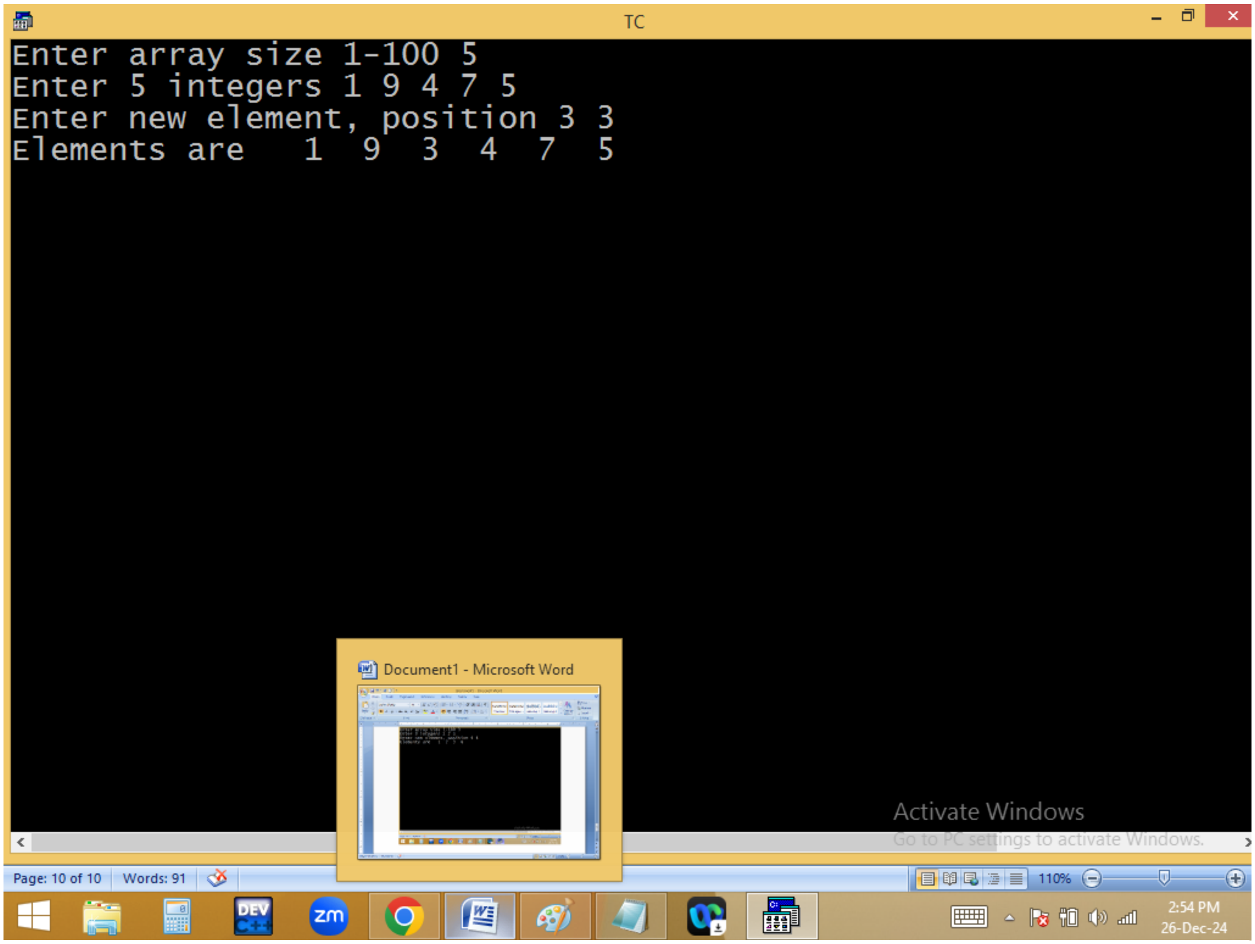

TC
Enter array size 1-100 3
Enter 3 integers 1 2 3
Enter new element, position 0 0
position 1 to 4 only
Activate Windows
Go to PC settings to activate Windows.
Page: 5 of 5
Words: 91
110%
2:52 PM
26-Dec-24


TC
Enter array size 1-100 3
Enter 3 integers 1 2 3
Enter new element, position 5 5
position 1 to 4 only_
Activate Windows
Go to PC settings to activate Windows.
Page: 6 of 6
Words: 91
110%
2:52 PM
26-Dec-24


TC
Enter array size 1-100 5
Enter 5 integers 1 2 3 4 5
Enter new element, position 9 9
position 1 to 6 only_
Activate Windows
Go to PC settings to activate Windows.
Page: 7 of 7
Words: 91
110%
2:53 PM
26-Dec-24


TC
Enter array size 1-100 3
Enter 3 integers 1 2 3
Enter new element, position 0 1
Elements are 0 1 2 3_
Activate Windows
Go to PC settings to activate Windows.
Page: 8 of 8
Words: 91
110%
2:53 PM
26-Dec-24


TC
Enter array size 1-100 3
Enter 3 integers 1 2 3
Enter new element, position 4 4
Elements are 1 2 3 4
Activate Windows
Go to PC settings to activate Windows.
Page: 9 of 9
Words: 91
110%
2:53 PM
26-Dec-24



2>=3

3>=3 a[3]=a[2]

4>=3 a[4]=a[3]

i=5; 5>=3; 5-- a[5]=a[4]

for(i=n; i>=pos;i--) a[i]=a[i-1];

3 >= 9

a[i]=newele;

2 9

2	5	~~6~~ 4	~~9~~ 6	~~10~~ 9	10			

0 1 2 → 3 → 4 → 5 6 7 8

newele = 4 pos=3 cell

n / ~~5~~ 6

a

z

n / 3 p / 0

i=3 >= 0

2>=0

1>=0

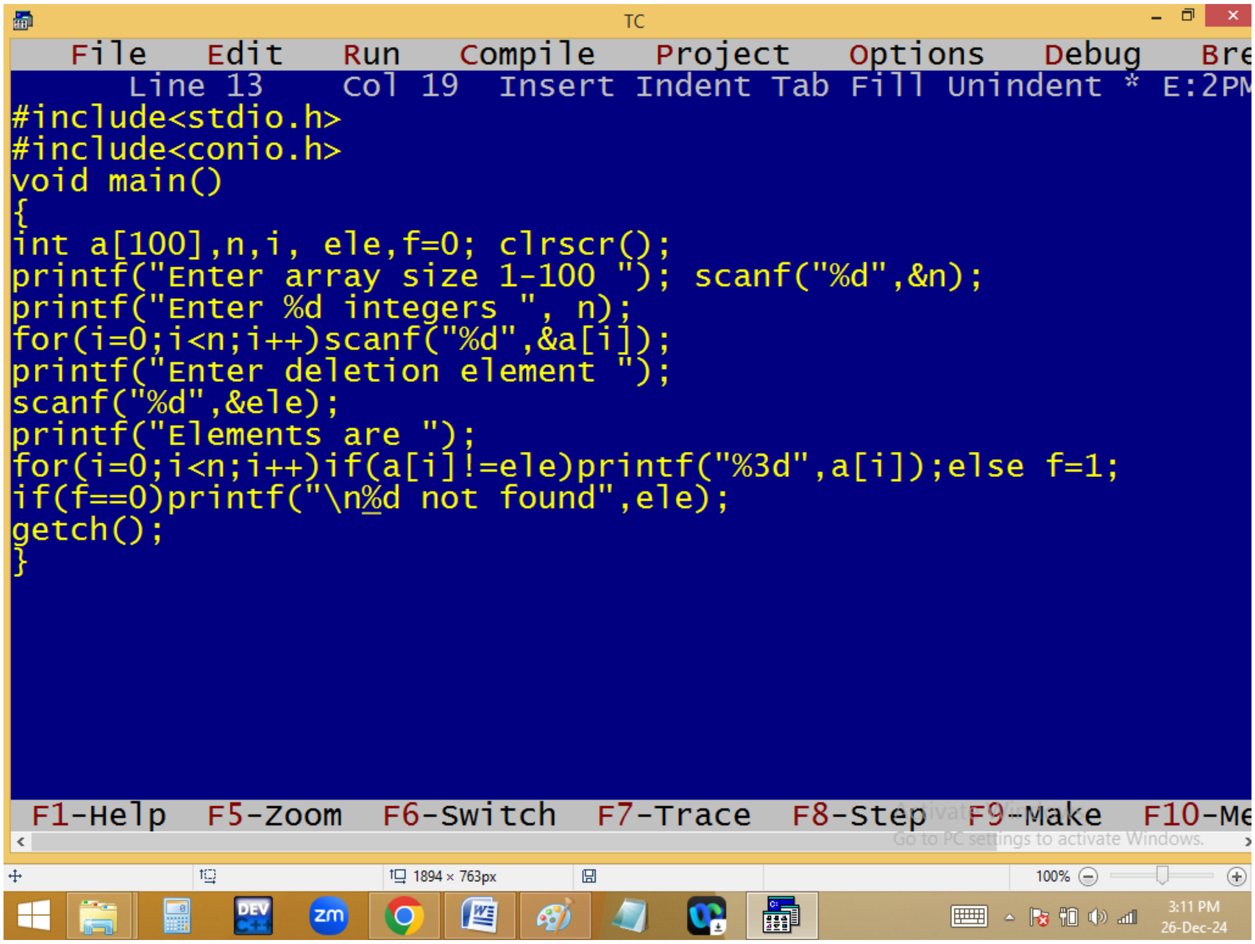
0 >= 0 ← -1

n / 3 p / 9

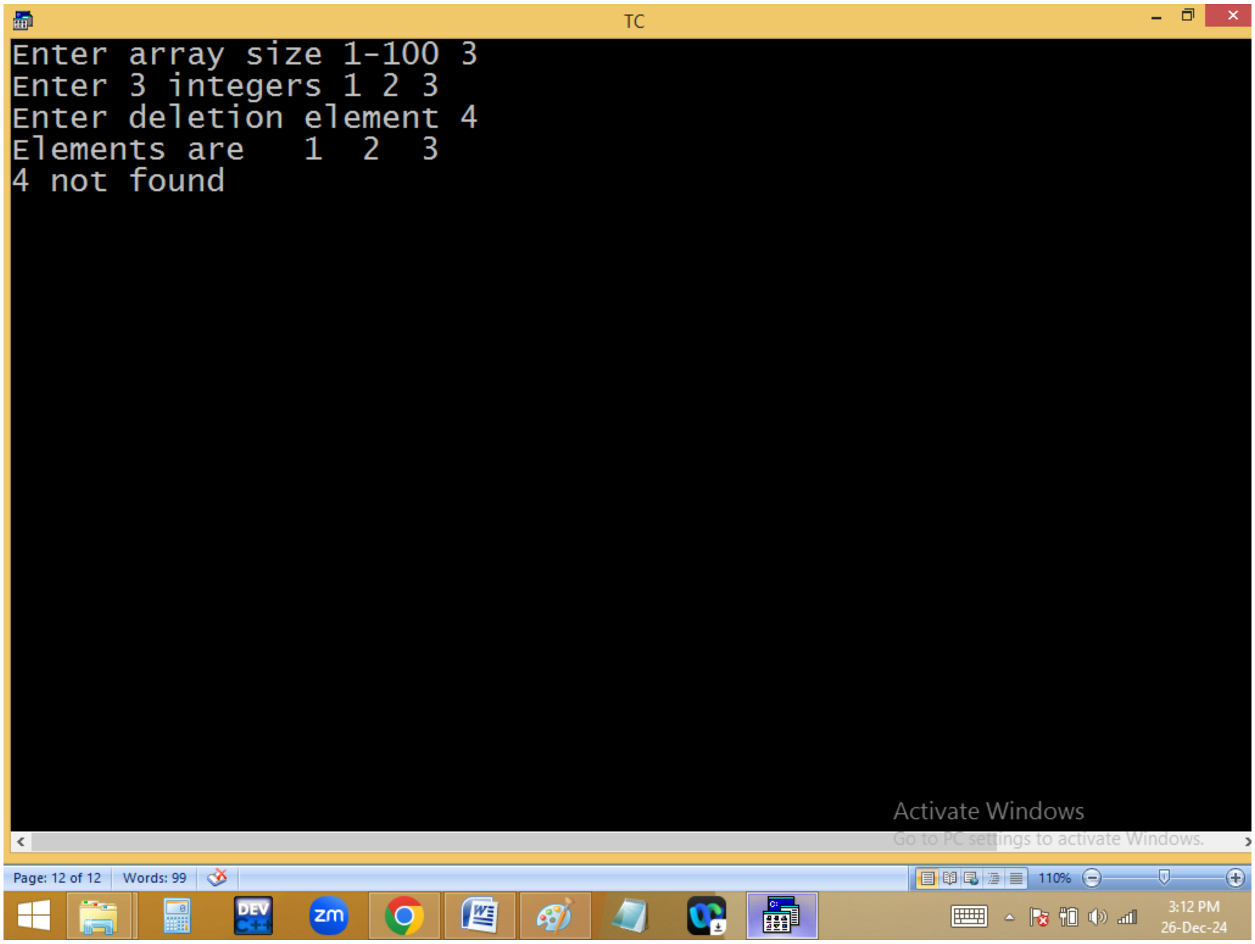
nani

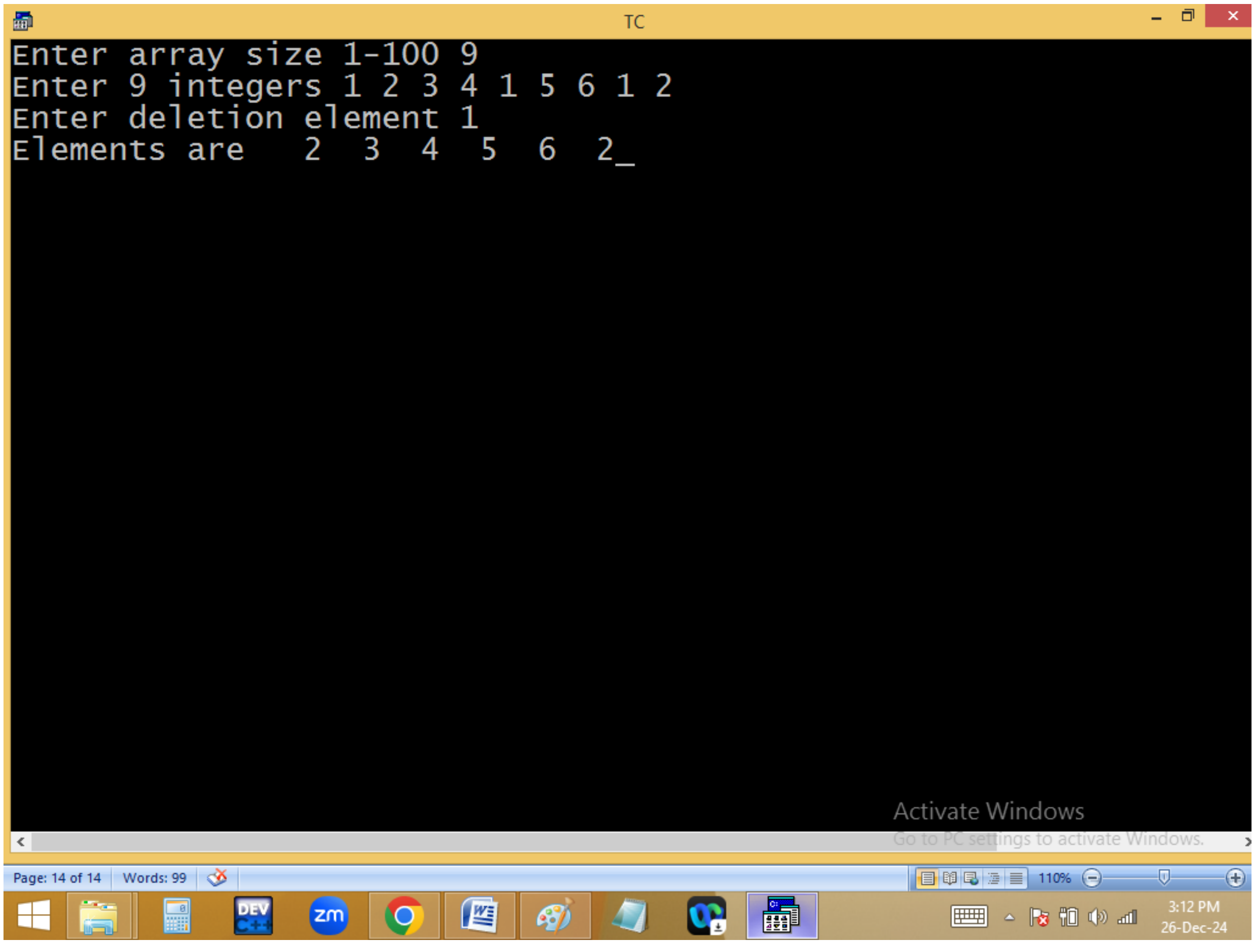
Deleting a particular element from array:

Temp method:



```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i, ele,f=0; clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter deletion element ");
scanf("%d",&ele);
printf("Elements are ");
for(i=0;i<n;i++)if(a[i]!=ele)printf("%3d",a[i]);else f=1;
if(f==0)printf("\n%d not found",ele);
getch();
}
```


TC
Enter array size 1-100 3
Enter 3 integers 1 2 3
Enter deletion element 4
Elements are   1   2   3
4 not found
Activate Windows
Go to PC settings to activate Windows.
Page: 12 of 12
Words: 99
110%
3:12 PM
26-Dec-24


TC
Enter array size 1-100 9
Enter 9 integers 1 2 3 4 1 5 6 1 2
Enter deletion element 1
Elements are 2 3 4 5 6 2_
Activate Windows
Go to PC settings to activate Windows.
Page: 14 of 14
Words: 99
110%
3:12 PM
26-Dec-24

2	5	7	1	4				
0	1	2	3	4	5	6	7	8

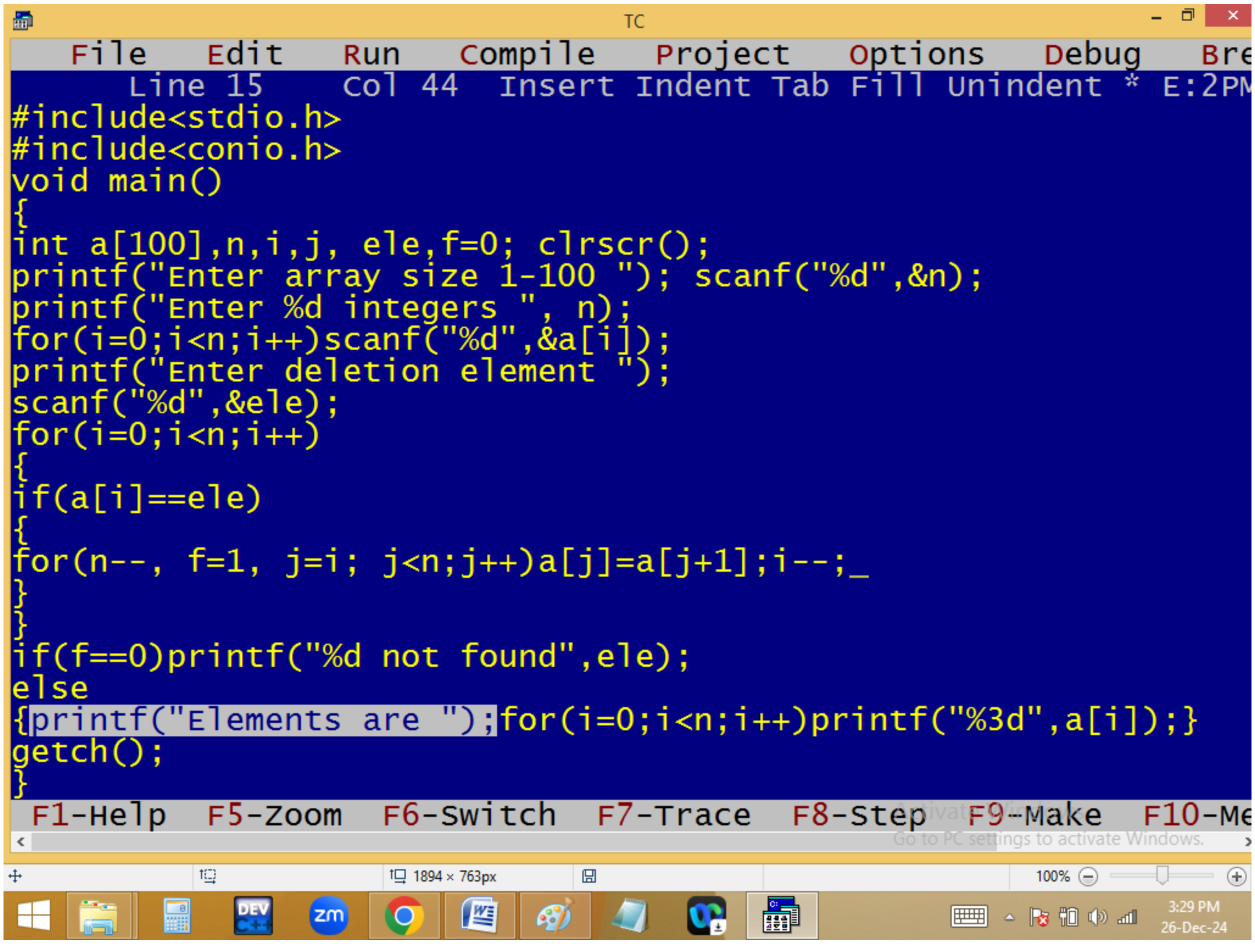
```
for(i=0;i<5;i++)
if( a[i]!=ele)p(a[i]);
```

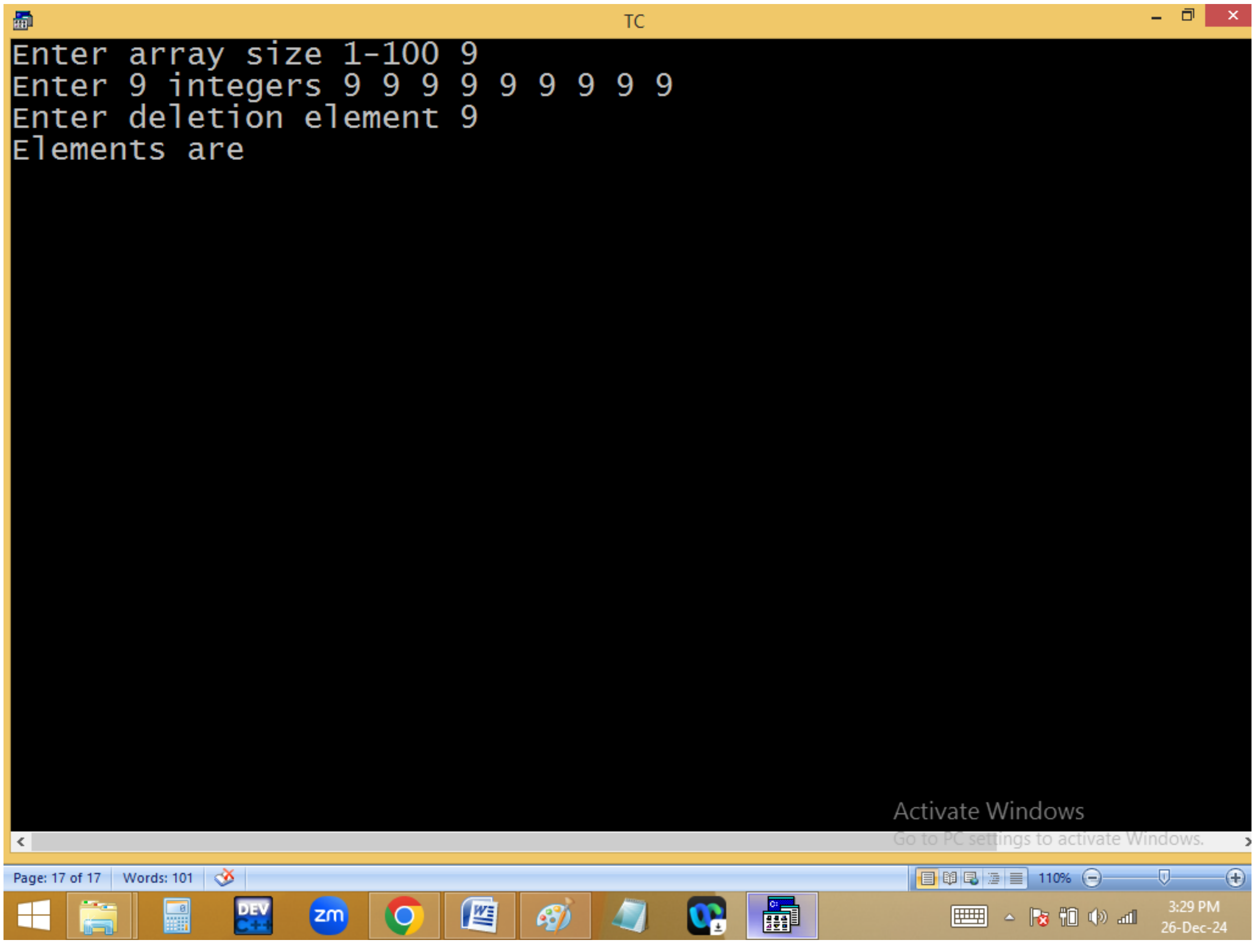
2 5 1 4

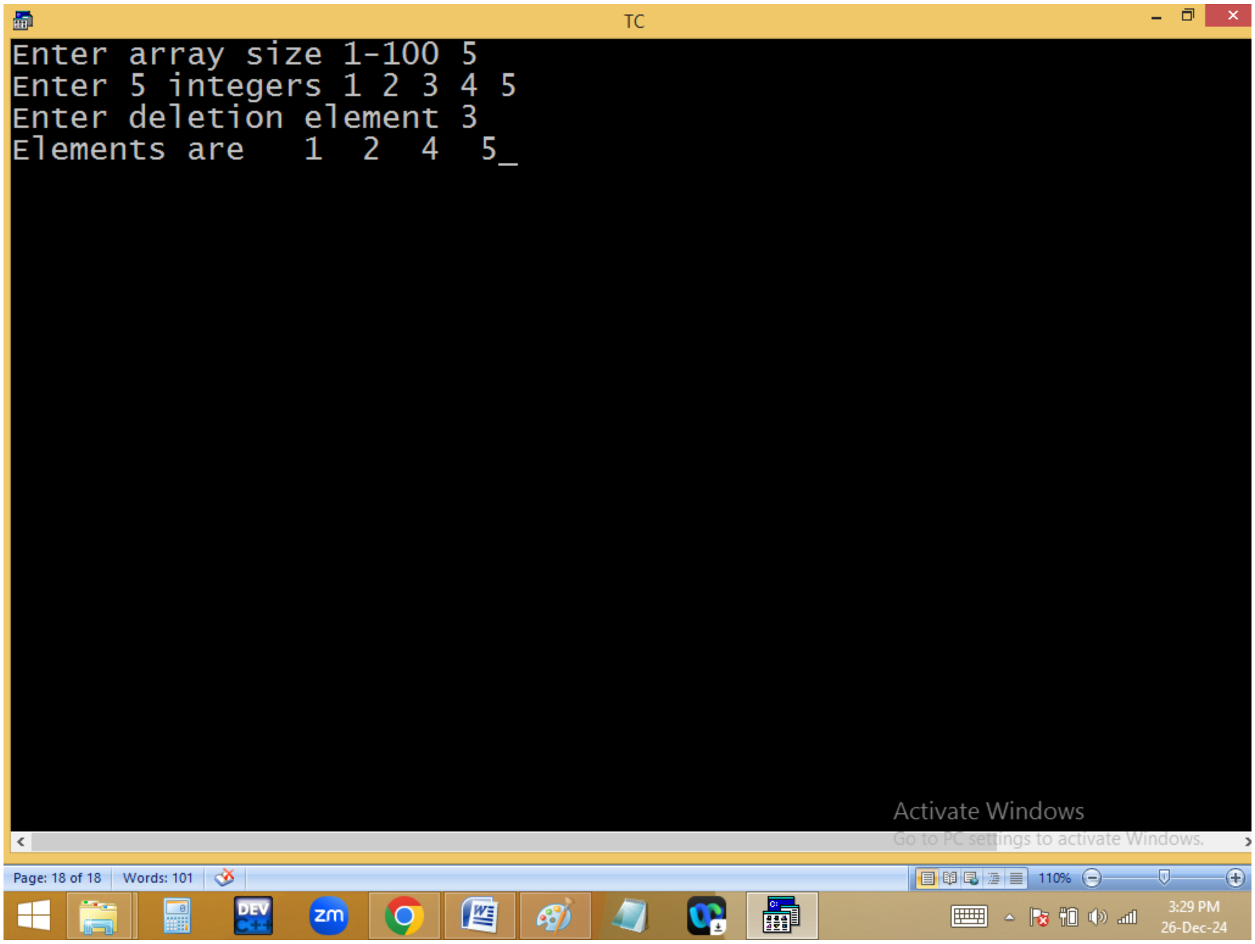
n: 5   ele: 7

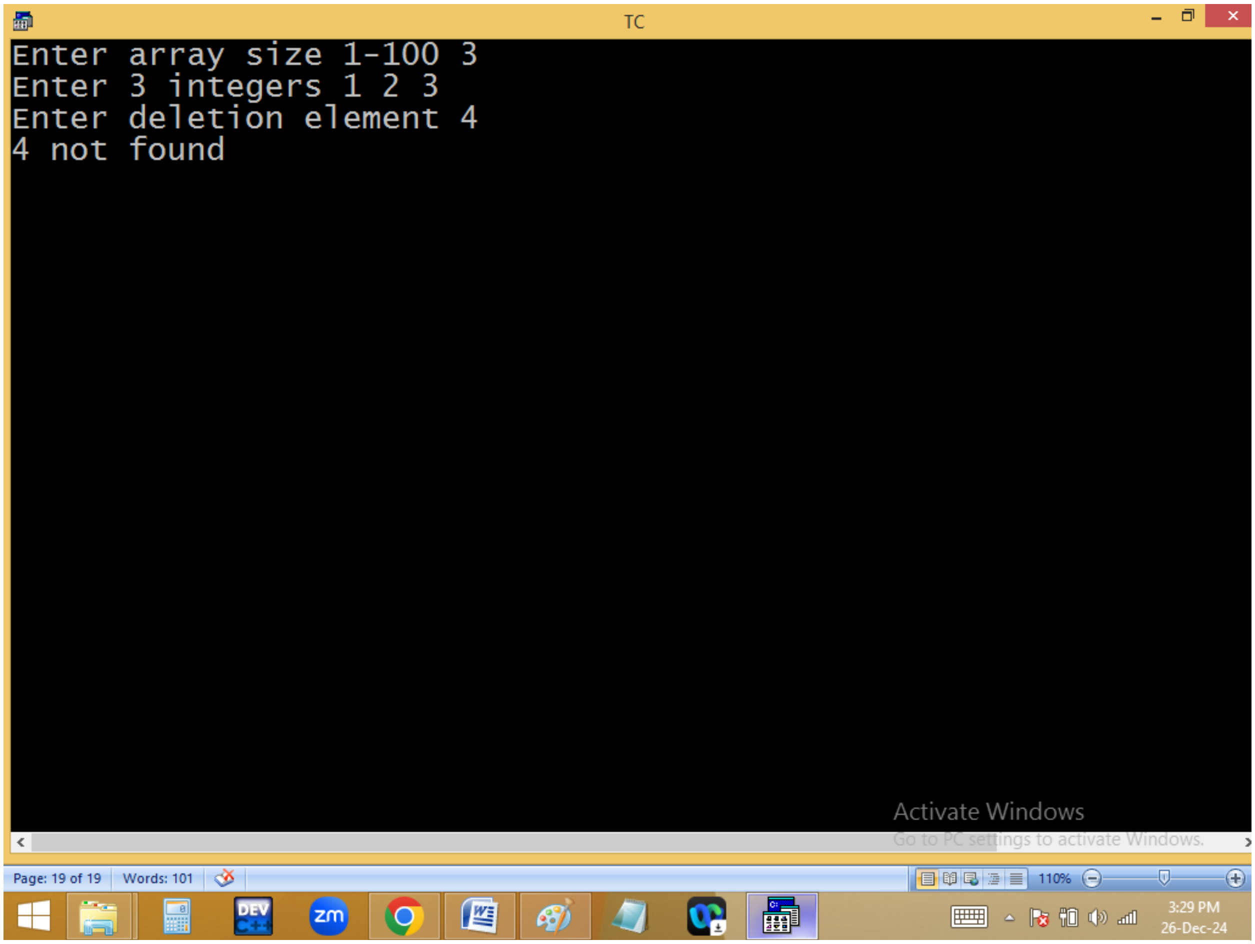
2 7
5
~~7~~
1
4

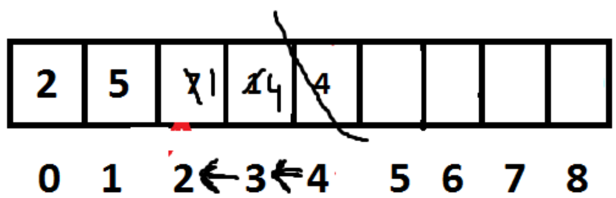
Permanent deletion:


TC
File Edit Run Compile Project Options Debug Bre
Line 15 Col 44 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j, ele,f=0; clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter deletion element ");
scanf("%d",&ele);
for(i=0;i<n;i++)
{
if(a[i]==ele)
{
for(n--, f=1, j=i; j<n;j++)a[j]=a[j+1];i--;_
}
}
if(f==0)printf("%d not found",ele);
else
{printf("Elements are ");for(i=0;i<n;i++)printf("%3d",a[i]);}
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.
1894 × 763px
100%
3:29 PM
26-Dec-24


TC
Enter array size 1-100 9
Enter 9 integers 9 9 9 9 9 9 9 9 9
Enter deletion element 9
Elements are
Activate Windows
Go to PC settings to activate Windows.
Page: 17 of 17
Words: 101
110%
3:29 PM
26-Dec-24


TC
Enter array size 1-100 5
Enter 5 integers 1 2 3 4 5
Enter deletion element 3
Elements are 1 2 4 5_
Activate Windows
Go to PC settings to activate Windows.
Page: 18 of 18
Words: 101
110%
3:29 PM
26-Dec-24



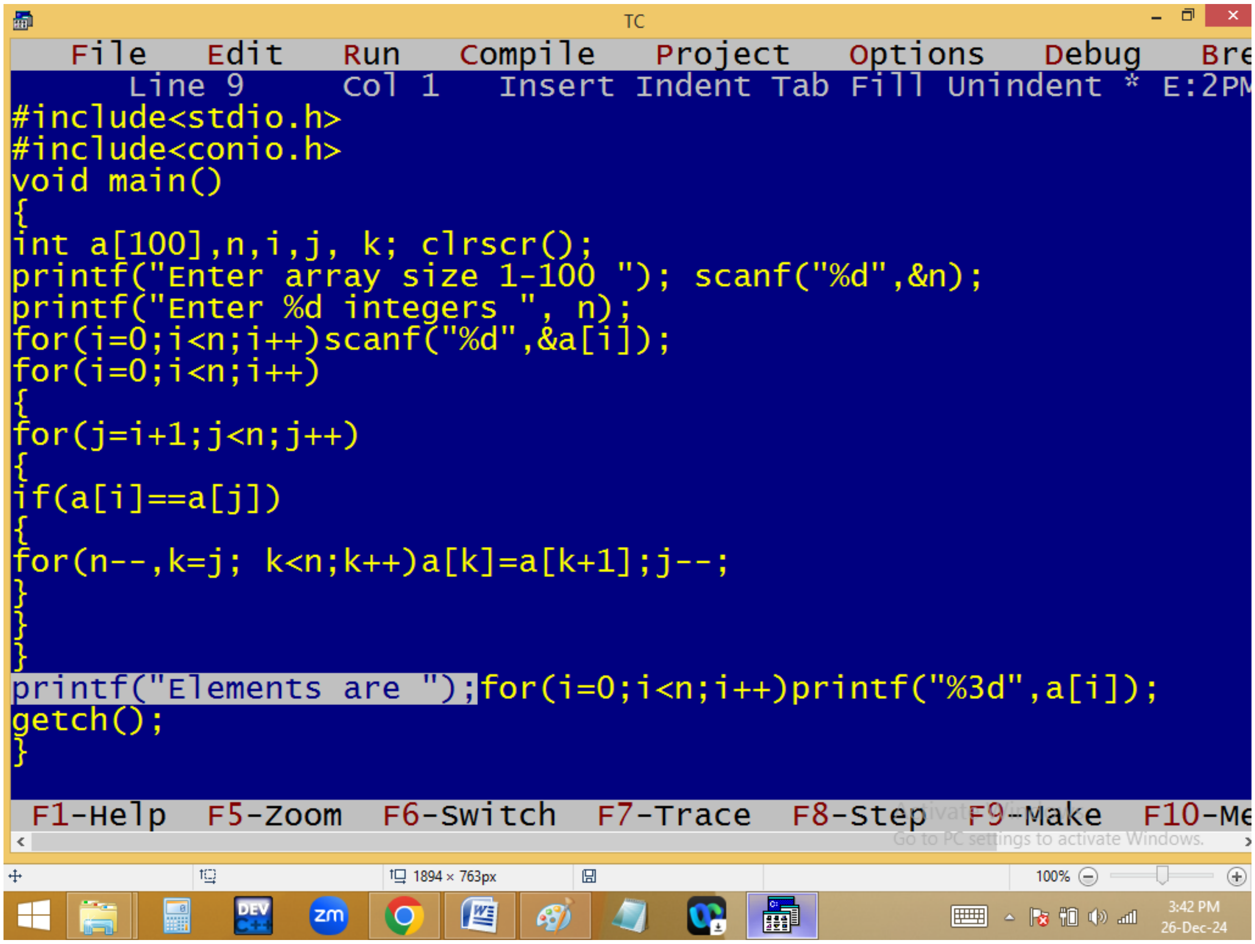


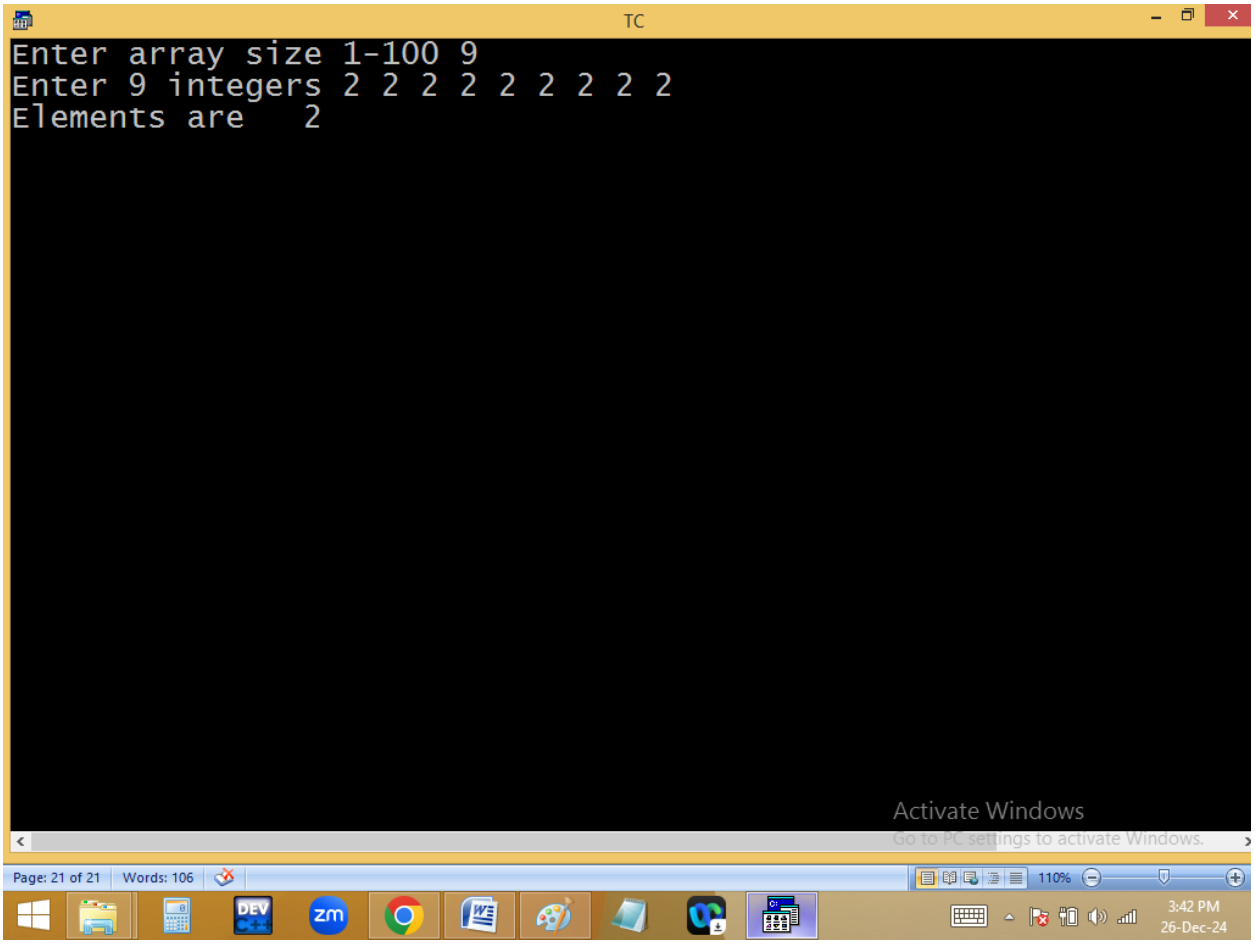

```
          5
for( i=0; i<n; i++)
{
if(a[i]==ele)
{
for(n--, f=1, j=i;j++)a[j]=a[j+1];
}
}
```

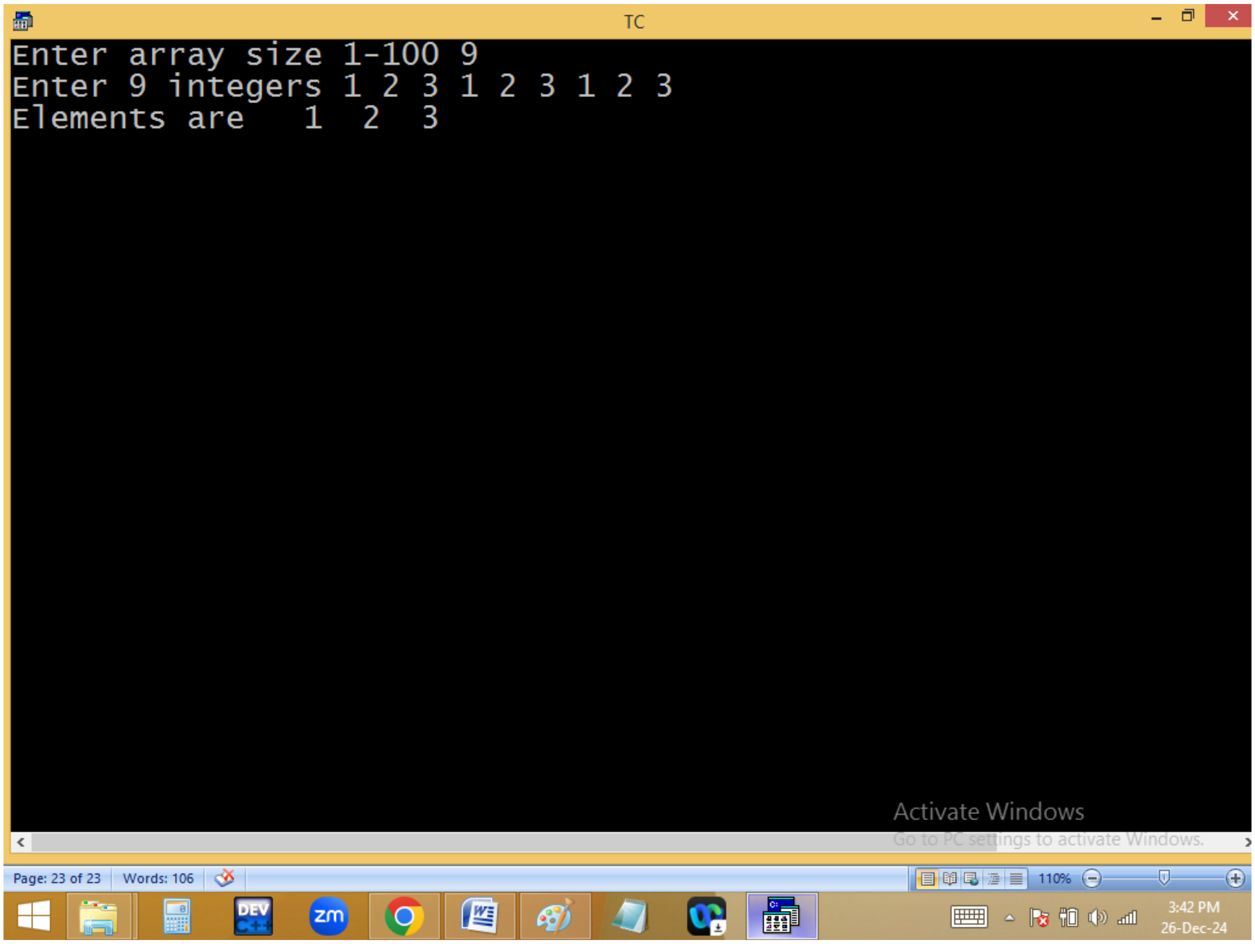
4<4

i	j	j+1	n	ele
0			~~5~~	7
1			4	
2	2 ← 3			
	3 ← 4			
	4			

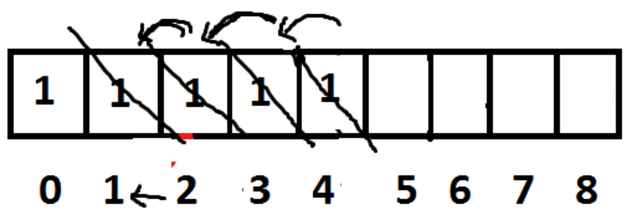
Deleting duplicate elements from array:


TC
File Edit Run Compile Project Options Debug Bre
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j, k; clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)
{
for(j=i+1;j<n;j++)
{
if(a[i]==a[j])
{
for(n--,k=j; k<n;k++)a[k]=a[k+1];j--;
}
}
}
printf("Elements are ");for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
1894 × 763px
100%
3:42 PM
26-Dec-24


TC
Enter array size 1-100 9
Enter 9 integers 2 2 2 2 2 2 2 2 2
Elements are   2
Activate Windows
Go to PC settings to activate Windows.
Page: 21 of 21
Words: 106
110%
3:42 PM
26-Dec-24




```
               5
for( i=0; i<n; i++)
{
for( j=i+1;j<n;j++)
{
if(a[i]==a[j])
{
for(n--, k=j;k<n;k++)a[k]=a[k+1]; j--;
}
}
}
```



n	i	j	k	k+1
~~5~~	0	1	1 ←	2
~~4~~		0	2 ←	3
~~3~~		1	3 ←	4
~~2~~		0	4	
1		1		

Home work:

```
               5
for( i=0; i<n; i++)
{

for( j=i+1;j<n;j++)
{
if(a[i]==a[j])
{
for(n--, k=j;k<n;k++)a[k]=a[k+1]; j--;
}
}
}
```

1	4	3	1	3	6	1	4	2
0	1←	2	3	4	5	6	7	8

n	i	j	k	k+1
~~5~~	0	1	1 ←	2
4			2 ←	3
			3 ←	4

1 - 3
4 - 2
3 - 2
6 - 1
2 - 1